

House Price Prediction — Project Summary

Objective

The goal of this project was to build and compare machine learning models that predict house prices based on property attributes such as area, number of bedrooms/bathrooms, stories, and amenities (air conditioning, parking, furnishing status, etc.).

Methodology

- Data Cleaning: Missing values were imputed (median for numeric, mode for categorical), duplicate rows removed, binary yes/no columns mapped to 1/0, and the multi-category furnishingstatus column one-hot encoded.
- Feature Set: 13 numeric features used for training after encoding (12 original + 2 furnishing dummies – 1 original categorical column).
- Train/Test Split: 80% training, 20% testing, with a fixed random seed for reproducibility.
- Models Trained: Linear Regression (baseline) and Random Forest Regressor (ensemble, 200 trees), evaluated using MAE, RMSE, and R² Score.

Key Findings

- Area showed the strongest positive correlation with price, followed by number of bathrooms and stories.
- Amenities such as air conditioning and a preferred location (prefarea) also contributed positively to price.
- Random Forest Regressor outperformed Linear Regression, achieving a higher R² score and lower RMSE/MAE, since it captures non-linear relationships and feature interactions that linear models miss.
- The Actual vs. Predicted scatter plot showed predictions clustering closely around the ideal diagonal line, with slightly higher variance for high-end, luxury-priced homes.

Conclusion

Tree-based ensemble methods such as Random Forest are better suited to this dataset than simple linear models, due to the mix of categorical and continuous features and non-linear price effects. Area, bathrooms, and key amenities are the most influential predictors of house price in this dataset.